

PROJECT REPORT

Introduction

TP53 is a tumor suppressor gene that encodes the p53 protein, often called the "Guardian of the genome". It plays a critical role in regulating the cell cycle, repairing DNA damage, and triggering cell death when the damage is beyond repair. Mutations in TP53 are found in more than 50% of all human cancers, making it one of the most significant genes in cancer biology.

Results

a) Sequence Statistic

Parameter	Value
Sequence ID	NM_000546.6
Sequence Length	2512 bp
GC Content	53.38%
ORFs Found	19
Longest ORF	ORF1 — Position 142-1324 1182 bp

b) ORF Annotation: A Total of 19 ORFs were identified with a minimum length of 100bp

ORF	Position	Length
ORF1 (Longest)	142–1324	1182 bp
ORF2	259–1324	1065 bp
ORF3	271–1324	1053 bp
ORF4	323–509	186 bp
ORF5	337–1324	987 bp
ORF6	538–1324	786 bp
ORF7	619–1324	705 bp
ORF8	646–1324	678 bp
ORF9	677–881	204 bp

ORF	Position	Length
ORF10	698–881	183 bp
ORF11	761–881	120 bp
ORF12	850–1324	474 bp
ORF13	868–1324	456 bp
ORF14	877–1324	447 bp
ORF15	1159–1324	165 bp
ORF16	2022–2499	477 bp
ORF17	2025–2499	474 bp
ORF18	2232–2499	267 bp
ORF19	2259–2499	240 bp

Interpretation:

The TP53 sequence (NM_000546.6) is 2512bp long with a GC Content of 53.38%, indicating a moderately GC-rich gene. A total of 19 ORFs were identified, of which ORF1(position 142-1324, 1182bp) is the longest and most likely represents the primary coding region. The remaining ORFs detected beyond position 2022 may represent regulatory elements. The high GC content overlaps with the primary ORF, indicating a strong biological significance.

Conclusions

Running this analysis helped me understand how raw sequence data can be computationally processed to extract meaningful biological information. It is necessary because analysing genes like TP53 computationally is essential, as manual analysis of thousands of base pairs is not feasible. Bioinformatics pipelines allow researchers to quickly extract sequence statistics, identify coding regions and connect findings to existing literature, all in an automated and reproducible way. In real research and clinical settings, this kind of analysis forms the foundation for understanding gene function and identifying mutations.